AI Provenance and Research Integrity Report

Mitigating Sycophancy in LLM Hypothesis Generation

I. Overview of Human-AI Collaboration

This report documents the collaborative workflow between the human researcher (Sarvagna Khatri) and the AI (Google Gemini 1.5 Pro / Gemini 3.5 Flash) during the empirical study: *Mitigating Sycophancy in LLM Hypothesis Generation: A Cost-Plausibility Analysis of RAG vs. Heterogeneous Multi-Agent Debate*.

The project utilized a "vibe coding" paradigm, where the human researcher directed high-level architectural decisions, experimental design, and critical verification, while the AI handled literature synthesis, boilerplate code generation, LaTeX formatting, and statistical execution.

II. Audit of Key AI-Assisted Activities

Activity 1: Experimental Design & Pipeline Engineering

- **Task:** Design a heterogeneous 2-agent debate loop with API error handling.
- **AI Proposal:** Generated the `debate_pipeline.py` script featuring a `while`-loop and exponential backoff to handle HTTP 503/429 errors.
- **Human Decision:** Dictated the specific adversarial personas (Proposer vs. Challenger) to forcibly break the "conformity" failure mode observed in prior literature.
- **Verification Method:** Executed the script locally on a Colab environment, monitored API token usage logs, and verified the backoff mechanism.
- **Outcome:** A stable, ungrounded multi-agent pipeline capable of tracking massive cumulative token costs.

Activity 2: LLM-as-a-Judge Evaluation Strategy

- **Task:** Evaluate the plausibility of the generated hypotheses.
- **AI Proposal:** Initially proposed a standard 1-10 grading prompt without strict formatting constraints.
- **Human Decision (AI Limitation):** Recognized that the RAG baseline scored suspiciously low (2.14) despite having the exact correct context, identifying a systemic "verbosity bias" in the judge.
- **Correction:** Explicitly directed the creation of an *anti-verbosity* judge prompt and forced a complete re-evaluation of the data.
- **Outcome:** A statistically defensible evaluation protocol yielding Wilcoxon $p < 0.001$.

Activity 3: Ablation Study Execution

- **Task:** Prove that the *heterogeneous* personas caused the plausibility increase, not just token scaling.
- **AI Proposal:** Acting as a peer reviewer, flagged the "Unproven Heterogeneous Claim" as a major vulnerability.
- **Human Decision:** Agreed with the critique and commissioned the `run_ablation.py` script to test a Homogeneous debate.
- **Verification Method:** Executed the ablation study on the frozen 50-sample split to check for "sycophancy collapse."
- **Outcome:** Empirical proof isolating the active variable (adversarial friction) required for high-quality generation.

III. AI Limitations and Manual Corrections

1. Hallucinated Parsing:
Observation: Early judge evaluations included conversational text alongside the 1-10 integer, breaking the evaluation script.
Correction: The human and AI collaborated to implement a robust Regex parser to forcibly extract the integer.

2. Blind Spot on Baseline Fairness:
Observation: The AI initially accepted the massive plausibility gap as a pure success story.
Correction: It required human reasoning to recognize the mathematical suspicion in RAG's low score, leading to manual anti-verbosity controls.

3. API Token Mathematics:
Observation: LLMs struggle to natively track their own token consumption across loops.
Correction: The human enforced the use of the `usage_metadata` attribute from the SDK to forcefully log tokens at every API call.

IV. Conclusions on Vibe Coding and Remaining Risks

What depended on human reasoning?
The core scientific integrity rested entirely on human oversight. The AI can write the code to loop the API, but the human must define *why* a heterogeneous persona matters and recognize when the LLM judge is biased.

Remaining Risks:
Parametric Leakage: The base model was likely exposed to the PubMedQA abstracts during pre-training.
Intra-Model Sycophancy: Using Gemini to judge Gemini introduces an unquantifiable preference bias.